

TASK 8

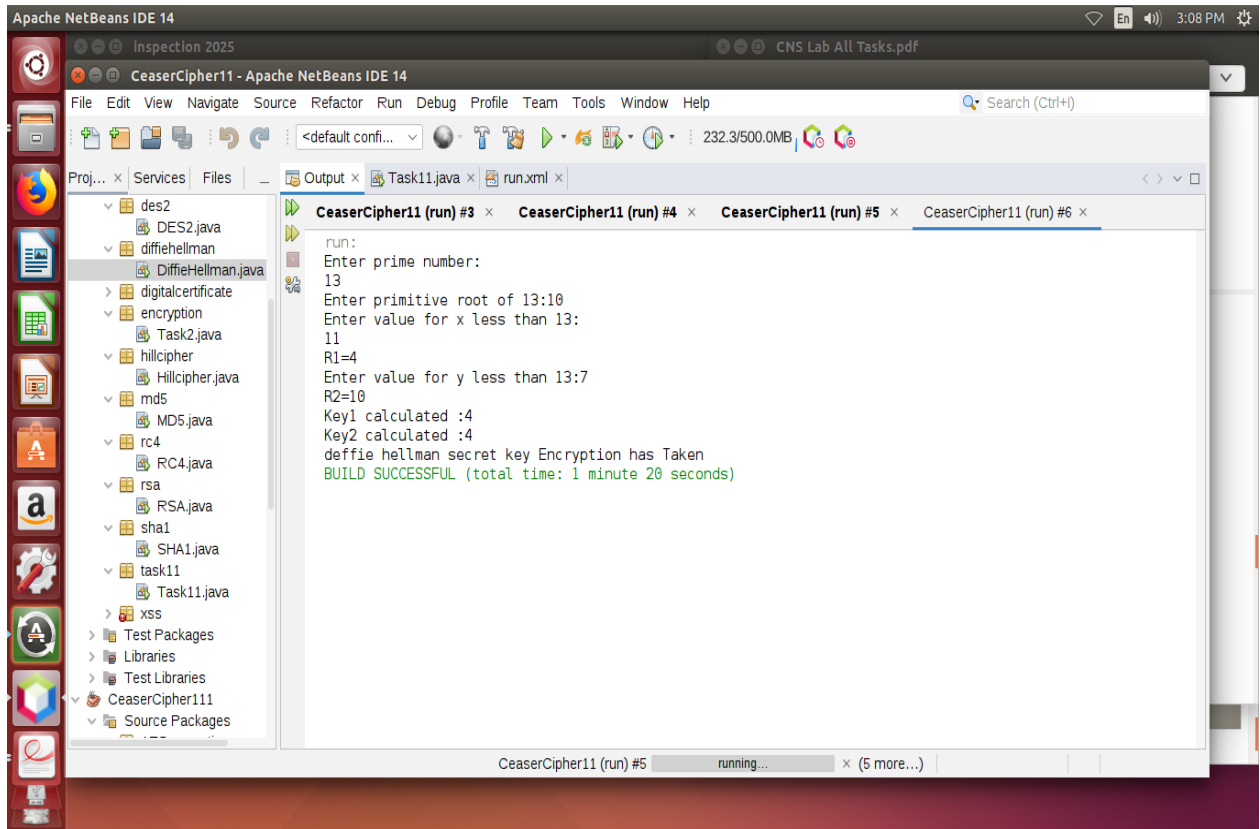
Implement key exchange protocol using the Diffie-Hellman algorithm

Aim: To implement the Diffie-Hellman Key Exchange mechanism.

Program:

```
package com.islab;
import java.io.*;
import java.math.BigInteger;
public class DiffieHellman
{
    public static void main(String[] args) throws IOException
    {
        BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
        System.out.println("Enter prime number:");
        BigInteger p=new BigInteger(br.readLine());
        System.out.print("Enter primitive root of "+p+":");
        BigInteger g=new BigInteger(br.readLine());
        System.out.println("Enter value for x less than "+p+":");
        BigInteger x=new BigInteger(br.readLine());
        BigInteger R1=g.modPow(x,p);
        System.out.println("R1="+R1);
        System.out.print("Enter value for y less than "+p+":");
        BigInteger y=new BigInteger(br.readLine());
        BigInteger R2=g.modPow(y,p);
        System.out.println("R2="+R2);
        BigInteger k1=R2.modPow(x,p);
        System.out.println("Key1 calculated :"+k1);
        BigInteger k2=R1.modPow(y,p);
        System.out.println("Key2 calculated :"+k2);
        System.out.println("deffie hellman secret key Encryption has Taken");
    }
}
```

OUTPUT:



```
run:  
Enter prime number:  
13  
Enter primitive root of 13:10  
Enter value for x less than 13:  
11  
R1=4  
Enter value for y less than 13:7  
R2=10  
Key1 calculated :4  
Key2 calculated :4  
deffie hellman secret key Encryption has Taken  
BUILD SUCCESSFUL (total time: 1 minute 20 seconds)
```

VIVA QUESTIONS

1. What is the Diffie-Hellman algorithm, and what problem does it solve?
2. How does the Diffie-Hellman key exchange process work?
3. Why is the Diffie-Hellman algorithm considered secure, and what mathematical concept underlies its security?
4. What are the main vulnerabilities of Diffie-Hellman, and how can they be mitigated?